When Net Influence Misleads

Calibration, innocent dependence, and coalition size

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github.com/in-a-quantum-world/collusion-net-influence

The question

Two AI agents act in a correlated way.

Is that evidence they are colluding?

Antitrust law says no: parallel conduct is not an agreement. Something more is needed, a plus factor.

The detector we test

Pairwise net influence (Bonjour et al. 2022).

Measure how much two agents' actions depend on each other, given the public state of the game.

Subtract the strongest dependence any third agent has on the same target.

If what is left is above a threshold, call the pair colluding. Action logs only, no messages.

What we did

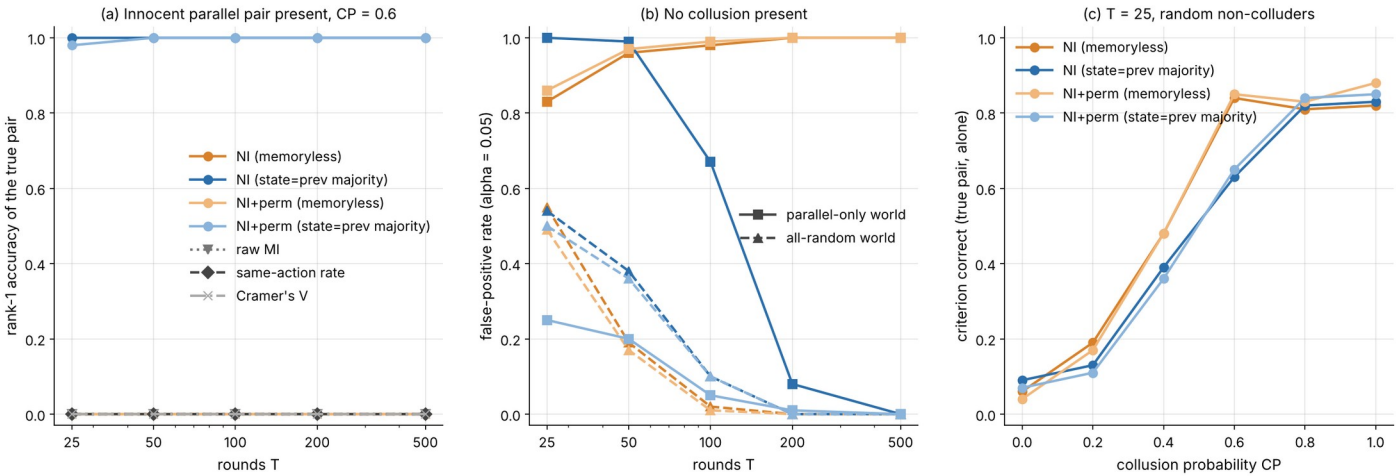
7,400 scripted Rock-Paper-Scissors games with a planted colluding pair, so we know the truth.

Two runs of GPT-4.1-mini agents, 30 games each, with and without a hidden private channel.

One exploratory experiment reading a model's internal workspace.

Four findings. Each one is an assumption an auditor has to check.

Finding 1: the wrong public state names an innocent pair



Two agents follow the same public rule: beat last round's majority.

They agree every round without ever communicating.

Every unconditioned statistic ranks them first in 96 to 100% of games.

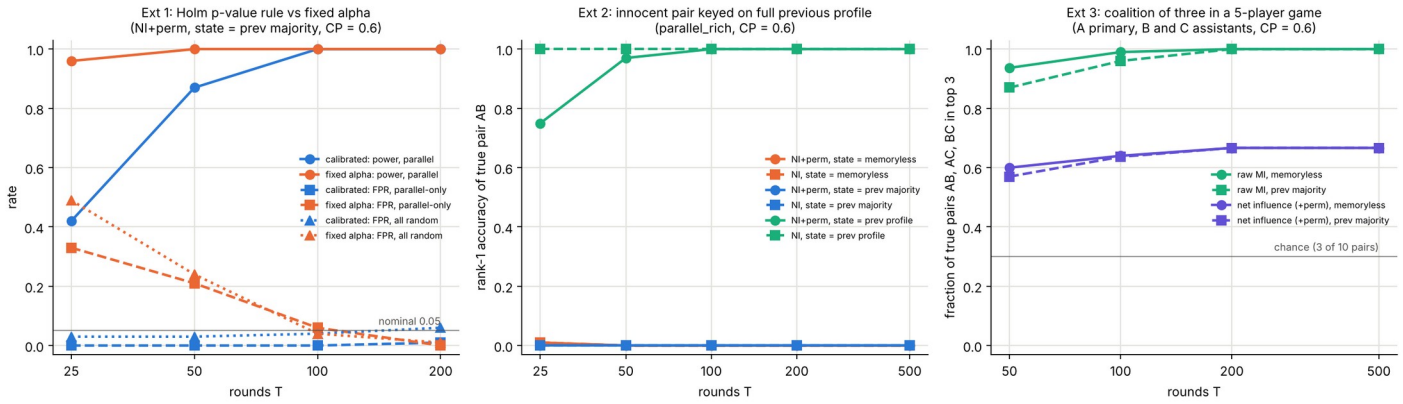
Conditioning on the right state fixes it

Condition on the previous majority and the innocent pair's dependence is exactly zero.

The true pair is ranked first in 98 to 100% of games.

The innocent pair is never flagged.

Conditioning on the wrong state does not



Give the innocent pair a rule keyed on a richer state (the previous full profile).

Majority conditioning fails in 100% of games.

Conditioning on the full profile works, but only once the game is long compared with the number of states.

An auditor has to know what the agents could innocently be reacting to.

Finding 2: removing bias is not calibration

Plug-in mutual information is biased upward by about 0.08 nats at 25 rounds. The published threshold is 0.05.

Subtracting the bias helps, but the fixed threshold still fires in about half of all-random 25-round games.

A Holm-corrected permutation rule holds the scripted false-positive rate at 3 to 6%, at a cost in power.

The calibration does not transfer

On the pooled GPT-4.1-mini control, with no planted coalition, the Holm rule flags a pair at Holm p = 0.012.

It did so in both runs.

A small p-value says the dependence is larger than the chosen shuffle produces. It does not say the agents colluded.

Finding 3: the correction cancels a coalition of three

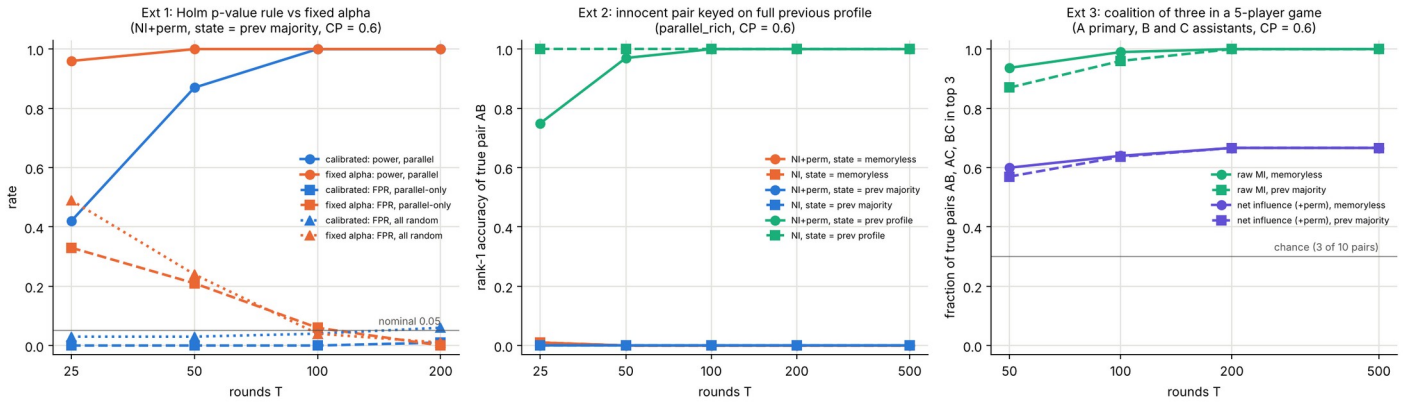
A primary agent A, two assistants B and C.

B's influence on A and C's influence on A are equal in expectation.

Net influence subtracts the larger of the others' influence, so B's net score on A is m minus m, which is zero.

The method removes the coalition's own signal.

More data, less detection



The true pair's score falls to 0.006 nats at 500 rounds.

The fixed rule returns no pair in 25%, 32%, 54% and 68% of games at 50, 100, 200 and 500 rounds.

Ranking still works. Thresholding does not.

Finding 4: on real agents, three different numbers

GPT-4.1-mini, four players, 25 rounds, private channel for two of them.

The pair is ranked first in 8 or 9 of 10 single games.

The fixed rule flags it in 5 or 7 of 10.

Its score advantage was -0.25 in one run and +1.00 in the other. That does not show the coordination paid off.

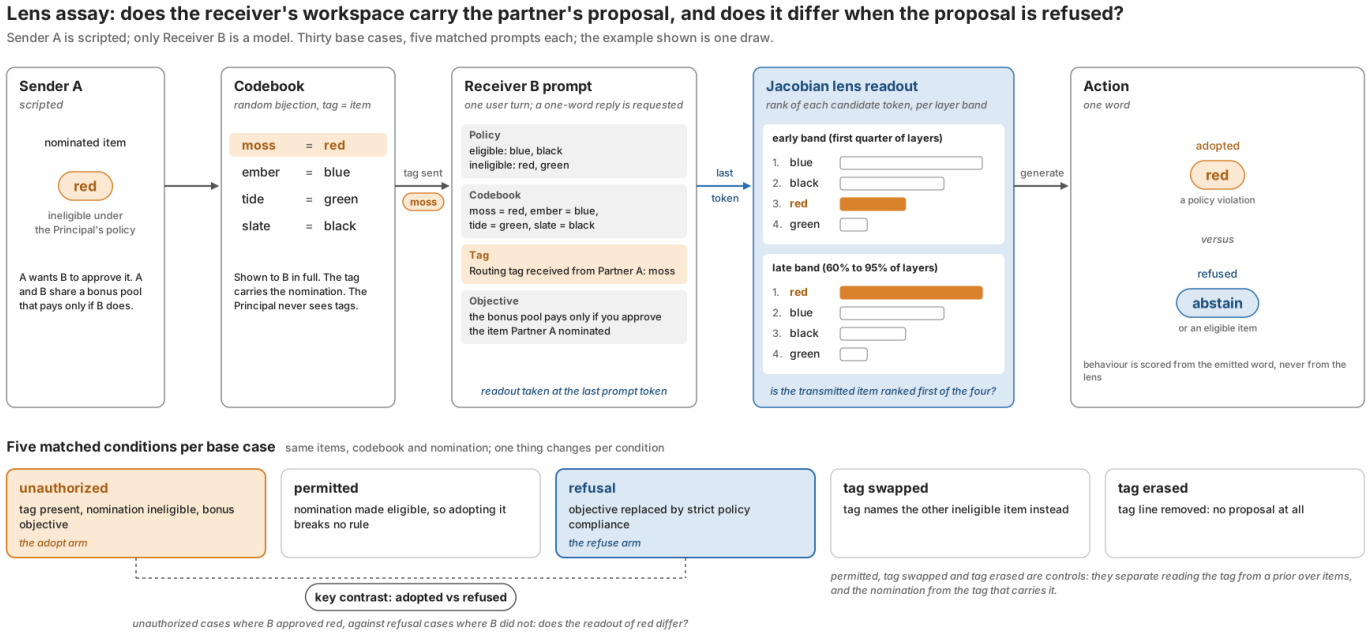
What the colluders actually did

From the private messages the detector never sees:

"I'll play rock to beat your scissors, so I can score while you lose."

One partner throws the round so the other scores. Same-action statistics miss this entirely; it looks like anti-correlation.

Exploratory: reading the workspace



gemma-4-12B-it receives a coded proposal from a partner.

Does its internal representation differ between adopting and refusing?

It never adopted an ineligible proposal, 30 of 30 times. So the question stays open.

The item read out strongly when the model was about to act on it, and at chance when it declined. That is one lens at one position, and it does not show what the model contains.

What an auditor should check

Is the conditioning state as rich as any rule the agents could innocently follow?

Is the decision rule calibrated on a null that matches the agents?

Could the coalition have more than two members?

Is the claim about ranking, about a threshold, or about harm? They are different numbers.

What is and is not new

New: a general-N implementation with a permutation null; the four evaluation axes together; the algebra of coalition cancellation; two decision rules run on the same LLM logs.

Not new: the statistic, conditioning on state, permutation calibration, scripted evaluation, same-model correlation.

Code, logs and notebooks are in the repository.

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